

Understanding Chunking Strategies

Do you need them? and how to choose the **right strategy** for your use case.



Do You Even Need to Chunk Your Data?

Chunking is not always necessary.

If your documents are **short** (emails, announcements, FAQs), you can often embed the whole thing.

But once text starts getting long, messy, or highly structured... chunking becomes essential for:

- Better retrieval
- Less hallucination
- More accurate answers
- Faster inference

So, the real question is: what kind of chunking does your data need?



Fixed-Size Chunking (The Simple & Fast One)

How it works: Splits text by tokens or characters.

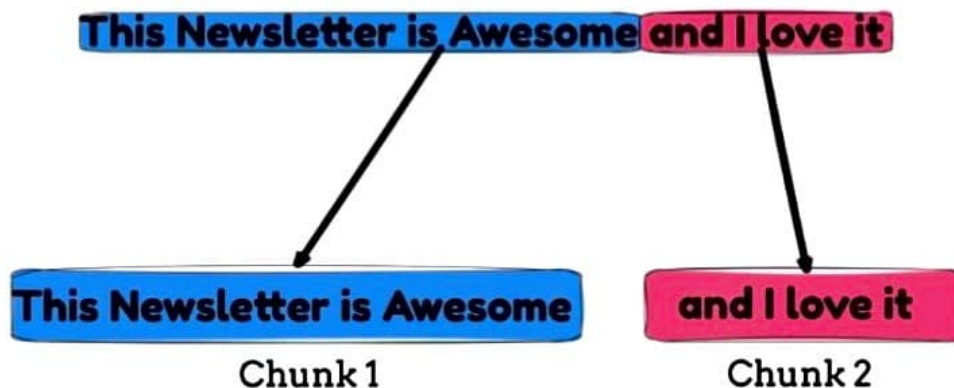
Best for: Short docs or when speed is everything.

Why use it:

- It's predictable
- It's easy to implement
- Works fine if your text has consistent flow

Examples: Emails, short blog posts, small FAQs.

When NOT to use: Documents with complex structure or shifting topics.



Recursive Chunking ("Smart Simple" One)

How it works: Splits text while trying to preserve some structure (headings, paragraphs).

Best for: Documents where some structure matters but you don't need a full semantic approach.

Why use it:

- More natural chunks
- Still efficient
- Reliable for medium-structured content

Examples: Product guides, research summaries

Recursive Character Text Splitting

Splitting the documents at fixed token length with overlaps. Chunks don't necessarily capture a single theme.

Hydroponics is an intelligent way to grow veggies indoors or in small spaces. In hydroponics, plants are grown without soil, using only a substrate and nutrient solution. The global population is rising fast, and there needs to be more space to produce food for everyone. Besides, transporting food for long distances involves lots of issues. You can grow leafy greens, herbs, tomatoes, and cucumbers with hydroponics.

thumarakesh.medium.com



Document-Based Chunking (The “As-Is” Method)

How it works: Splits strictly by document boundaries—sections, headers, pages.

Best for: Highly structured documents.

Why use it:

- Perfect when format already tells you how to chunk
- No guesswork

Examples: News articles, tickets, Markdown docs.

Heading This is a heading. --- ## Subheading This is a subheading. We can continue with more content here.



Semantic Chunking (The “Intelligent” One)

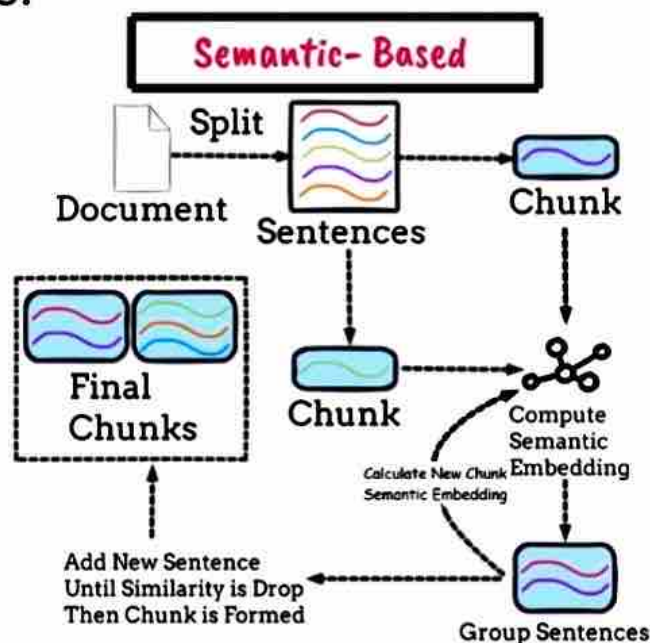
How it works: Splits text by meaning—topics, ideas, concept shifts.

Best for: Dense, technical, or narrative documents.

Why use it:

- Chunks follow human-readable meaning
- Great for Q&A
- Reduces retrieval of irrelevant context

Examples: Scientific papers, textbooks, whitepapers.



LLM-Based Chunking (Meaning + Context” One)

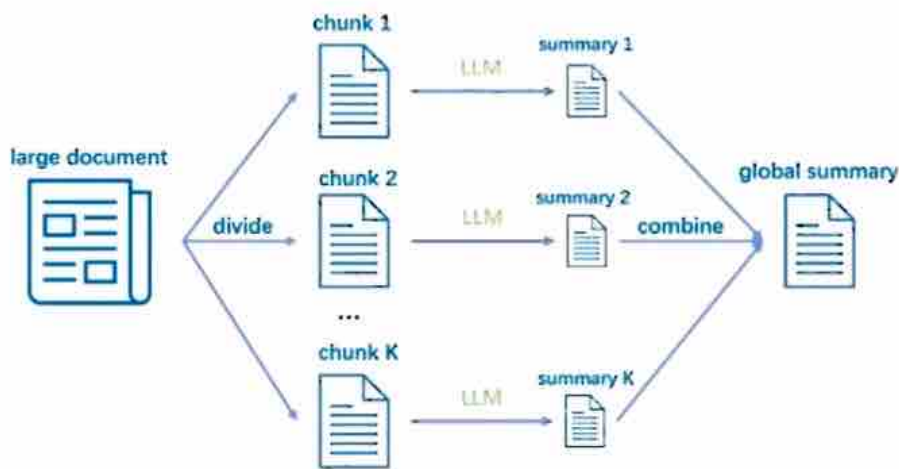
How it works: LLM decides where to split based on context.

Best for: Long, nuanced docs where fixed rules break.

Why use it:

- Highest accuracy in meaning boundaries
- Particularly strong for reasoning tasks

Examples: Legal opinions, long reports, medical records.



Agentic Chunking (The “Custom Strategy” One)

How it works: An AI agent analyzes the text and designs the chunking approach.

Best for: Complex, messy documents that need adaptive logic.

Why use it:

- Fully flexible
- Handles special formatting and unique structures

Examples: Regulatory filings, multi-section contracts, policies.



Late Chunking (The “Embed First, Split Later” Approach)

How it works: Embed the entire document → then derive chunk embeddings.

Best for: When you need chunks to know the whole story.

Why use it:

- Fixes “isolated chunk” issues
- Improves retrieval on interconnected ideas

Examples: Case studies, long-form reports, manuals.



Hierarchical Chunking (The “Multi-Level” Method)

How it works: Splits by sections → paragraphs → sentences.

Best for: Large, structured docs with both summary + detail needs.

Why use it:

- Flexible retrieval (high-level or deep dive)
- Great for regulation-heavy text

Examples: Handbooks, government regulations, software docs.

